

Part III

Electromagnetism

- Cavendish began his experiments on electrostatics in 1771.
- Coulomb published his famous result in 1789.
- Fifty years later, Faraday discovered the law of electromagnetic induction.
- In 1865, Maxwell summarized the laws of the motion of the electromagnetic field into the Maxwell equations and predicated the existence of electromagnetic waves. Hertz confirmed the existence of electromagnetic waves by experiments.

The classical electromagnetism is one of greatest achievement in Physics.

- Its wide application in engineering made the modern civilization possible.
- The theory of electromagnetism has the most ‘beautiful’ mathematical form.

Maxwell equations printed on T-shirts

self-consistent simple and brief symmetric

harmonic

Although the theory of electromagnetism is
‘beautiful’ as well as powerful, it has the limitation.

It cannot be used to describe the electromagnetic
interaction among elementary particles.

The quantum electrodynamics should be used in
that case. That’s why the what we will study in this
part called classical electromagnetism.

Chapter 15 Electrostatic Field

15.1 Electric Charge and Coulomb's Law

Electric charge —an intrinsic property of matters

gravitational mass — source of gravitational force

electric charge —source of electrostatic force

Two kinds of electrification: positive and negative

glass rod rubbing with silk brings **positive** charge;

plastic rod rubbing with fur brings **negative** charge

In Section 29.5, the electrification is defined on basic level through decay process of particle.

Some facts about electric charges:

- all charges are equivalent, and algebraically additive.
- Like charges repel, unlike charges attract.
- Electricity neutrality of objects is the most common occurrence.

Quantization of charge

Experiments show that charges are **quantized**, i.e., it cannot be subdivided continuously. There is a **fundamental charge**. The value is

$$e = 1.60217733(49) \times 10^{-19} \text{C} \quad 1\text{C} = 1\text{A} \cdot \text{s}$$

Is e really fundamental? fractional charges?

up quark	down quark	strange quark
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$2/3$	$-1/3$	$-1/3$	e
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Confinement of quark

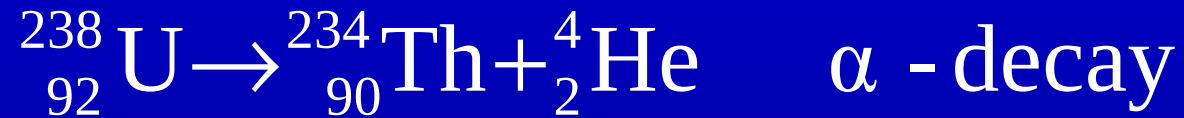
Theory: fractional charges of the collective excitations in polymer

Experiments: B. Fairbank et al (1965, 1977~1981) found that on superconducting niobium ball of diameters $280\mu\text{m}$, masses 10^{-7}kg , there exist fractional charges $\pm e / 3$

1909 Millikan wrote: “...I have discarded one uncertain and unduplicated observation apparently upon a singly charge drop, which gave a value of the charge on the drop some 30 per cent lower than the final value of e .”

Conservation of charge

Charge is **conserved** in large-scale events or on the atomic and nuclear level.



$$e^{-} + e^{+} \rightarrow 2\gamma \quad \text{pair - annihilation}$$

$$n \rightarrow p + e + \bar{\nu}_e \quad \beta - \text{decay}$$

Particle-antiparticle; Charge conjugate symmetry

Whatever has **charge** has **mass**. In other words, a particle of zero rest mass has zero charge.

Coulomb's law

Force acted on q_2
by q_1

$$\mathbf{F}_{21} = k \frac{q_1 q_2}{r_{12}^3} \mathbf{r}_{12} = -\mathbf{F}_{12}$$

$$k = \frac{1}{4\pi\epsilon_0} \approx 9 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2}$$

two-body action; action-at-a-distance;
inverse-square law

$$\frac{1}{r^{2+\epsilon}} \quad \epsilon \leq (2.7 \pm 3.1) \times 10^{-16}$$

Example 15.1 In the **Bohr model** of hydrogen atom

$$\langle r \rangle_{e-p} = 0.53 \text{Å}$$

$$\begin{aligned} F_e &= \frac{1}{4\pi\epsilon_0} \frac{e^2}{\langle r \rangle^2} \\ &= 9 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2} \frac{(1.6 \times 10^{-19} \text{C})^2}{(0.53 \times 10^{-10} \text{m})^2} \\ &= 8.1 \times 10^{-8} \text{N} \end{aligned}$$

$$\begin{aligned} F_g &= G \frac{m_e m_p}{\langle r \rangle^2} \\ &= 6.7 \times 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2} \frac{(9.1 \times 10^{-31} \text{kg})(1.7 \times 10^{-27} \text{kg})}{(0.53 \times 10^{-10} \text{m})^2} \\ &= 3.7 \times 10^{-47} \text{N} \end{aligned}$$

$$\frac{F_e}{F_g} \propto 10^{39}$$

Coulomb interaction dominants.

Chemical processes are due to the electrical interaction among atoms and molecules.

In a nucleus

Z (atomic number) + N (neutron number)

= A (mass number)

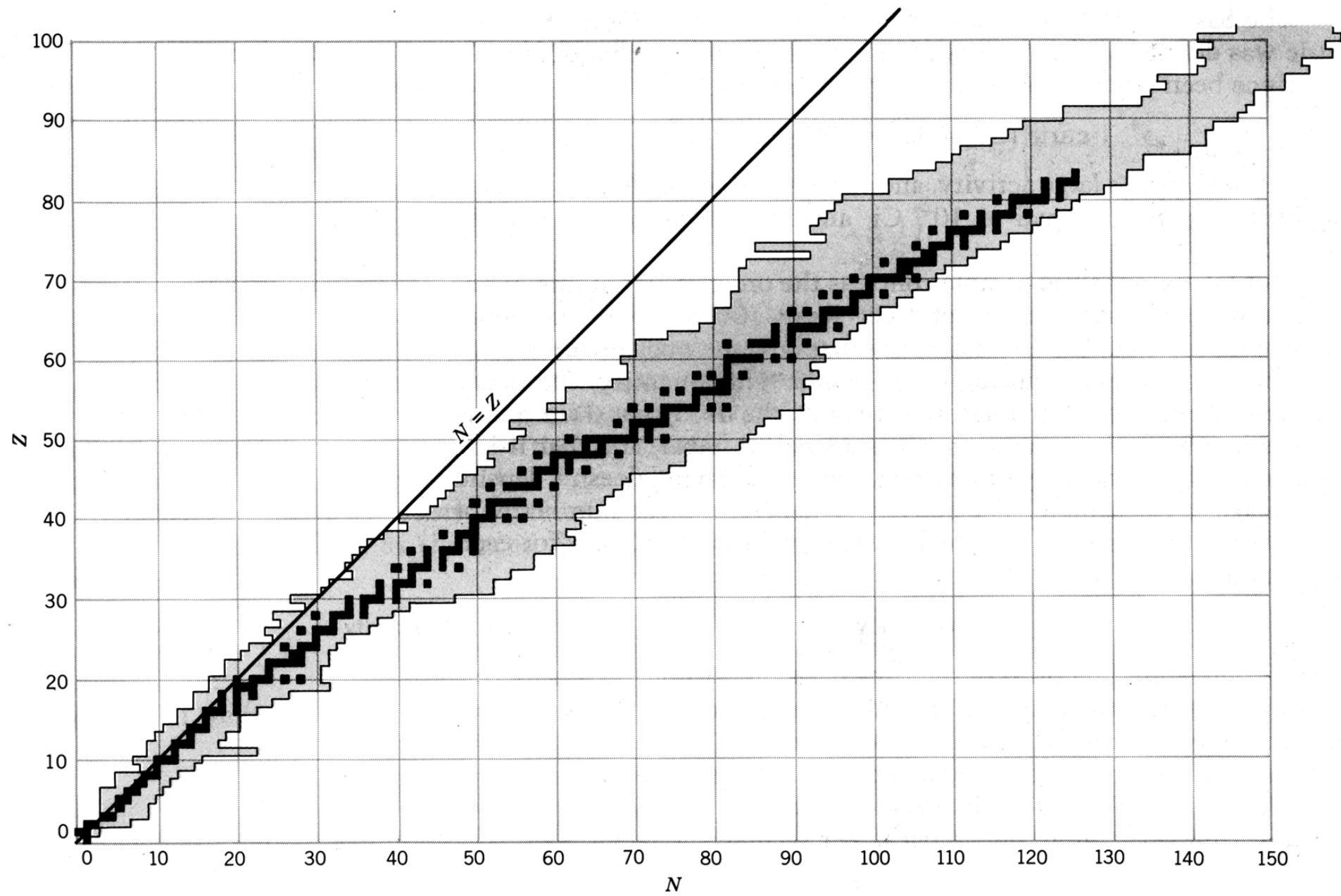
$$r \propto A^{1/3} \times 10^{-15} \text{ m}$$

$$\langle r \rangle_{\text{p-p}} \approx 4.0 \times 10^{-15} \text{ m}$$

$$F = \frac{(9 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(1.6 \times 10^{-19} \text{ C})^2}{(4.0 \times 10^{-15} \text{ m})^2} = 14 \text{ N}$$

How can protons stay together to form a nucleus?

There exists a strong, short ranged attractive interaction-**nuclear force** to counterbalance the Coulomb repulsion.



The neutrons contribute to the attractive nuclear force without adding electrical repulsion. They produce a stabilizing effect.

The nuclear fission and the α -decay are due to the Coulomb repulsion.

15.2 Electrostatic field

Electric Field is defined operationally as

$$\mathbf{E} = \lim_{q_0 \rightarrow 0} \frac{\mathbf{F}}{q_0}$$

Field around a point charge

$$\mathbf{E}_q = k \frac{qq_0}{r^3} \mathbf{r} \times \frac{1}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \mathbf{e}_r$$

Lines of force - the lines that at each point on them is tangent to the electric field at the point.

The differential equation of the lines of the force is

$$\frac{dx}{E_x} = \frac{dy}{E_y} = \frac{dz}{E_z}.$$

Example 15.2 Find the equations of the lines of the force for the electric field of a single point charge.

Solution: The electric field around a single point charge is

$$\vec{E} = \frac{q}{4\pi\epsilon_0} \frac{x\vec{i} + y\vec{j} + z\vec{k}}{r^3}.$$

Then, on the x - y plane, the differential equation for the lines of force is

$$\frac{dy}{dx} = \frac{E_y}{E_x} = \frac{y}{x}.$$

The solution of the equation is

$$y = kx$$

which represents a family of straight lines radiating from the point charge.

Principle of superposition

The total force acted on a charge q by a set of charges $q_1, q_2 \dots q_n$ is

$$\vec{F} = \sum_{i=1}^n \vec{F}_i = \sum_{i=1}^n \frac{qq_i}{4\pi\epsilon_0} \frac{\vec{r}_i}{r_i^3}$$

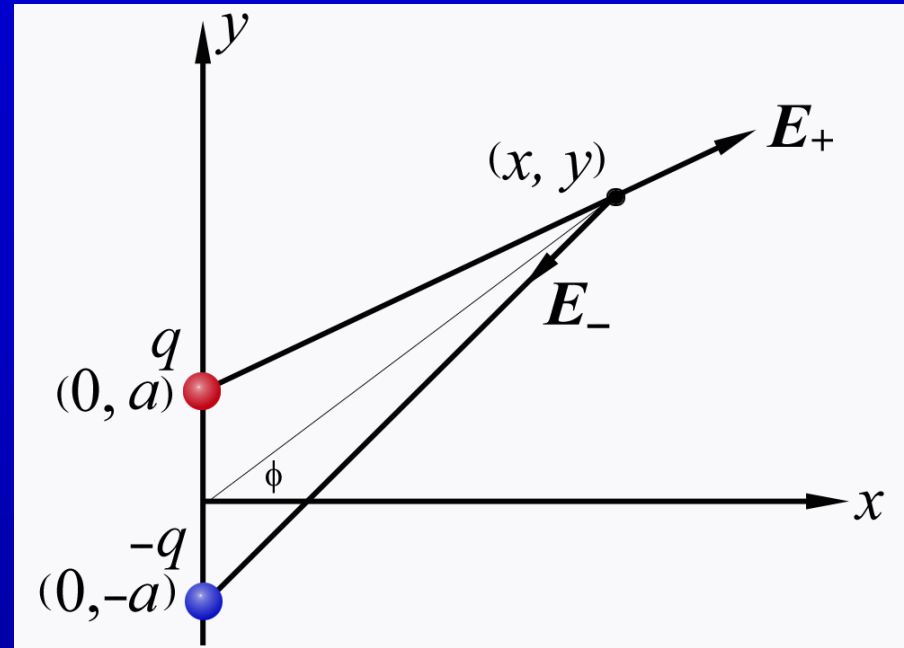
where $\vec{r}_i$ is the displacement vector from q_i to q .

Then, the resultant electric field is

$$\vec{E} = \sum_{i=1}^n \vec{E}_i = \sum_{i=1}^n \frac{q_i}{4\pi\epsilon_0} \frac{\vec{r}_i}{r_i^3}$$

Field around a dipole

$$\mathbf{E} = E_x \mathbf{i} + E_y \mathbf{j}$$



$$E_x = \frac{1}{4\pi\epsilon_0} \left\{ \frac{qx}{[x^2 + (y-a)^2]^{3/2}} + \frac{-qx}{[x^2 + (y+a)^2]^{3/2}} \right\}$$

$$E_y = \frac{1}{4\pi\epsilon_0} \left\{ \frac{q(y-a)}{[x^2 + (y-a)^2]^{3/2}} + \frac{-q(y+a)}{[x^2 + (y+a)^2]^{3/2}} \right\}$$

For far zone ($r \gg a$), or far field points

$$E_x \approx \frac{1}{4\pi\epsilon_0} \frac{3xy}{(x^2 + y^2)^{5/2}} (q \cdot 2a)$$

$$E_y \approx \frac{1}{4\pi\epsilon_0} \frac{2y^2 - x^2}{(x^2 + y^2)^{5/2}} (q \cdot 2a)$$

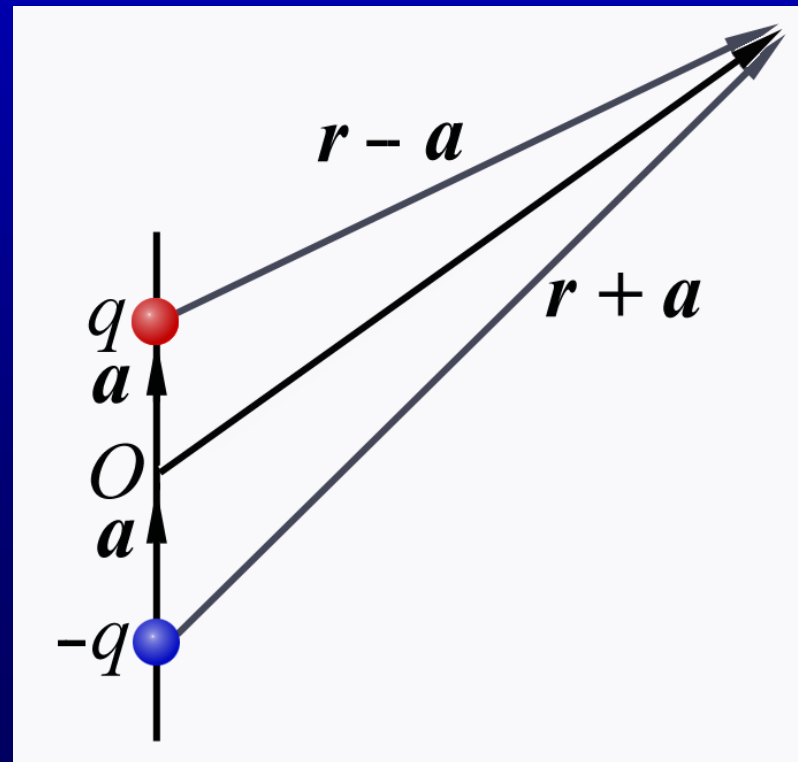
$$\frac{3xy}{(x^2 + y^2)^{5/2}} = \frac{3}{2r^3} \sin 2\varphi \sim \frac{1}{r^3}$$

$$\frac{2y^2 - x^2}{(x^2 + y^2)^{5/2}} = \frac{1}{2r^3} (1 - 3\cos 2\varphi) \sim \frac{1}{r^3}$$

$$E \sim \frac{q}{r^2} \left(\frac{a}{r} \right)^1$$

Point charge

$$\begin{aligned}
 \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\mathbf{r} - \mathbf{a}|^3} (\mathbf{r} - \mathbf{a}) - \frac{q}{|\mathbf{r} + \mathbf{a}|^3} (\mathbf{r} + \mathbf{a}) \right] \\
 &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \left\{ \frac{\mathbf{r}}{r} \left[\frac{r^3}{|\mathbf{r} - \mathbf{a}|^3} - \frac{r^3}{|\mathbf{r} + \mathbf{a}|^3} \right] - \frac{\mathbf{a}}{r} \left[\frac{r^3}{|\mathbf{r} - \mathbf{a}|^3} + \frac{r^3}{|\mathbf{r} + \mathbf{a}|^3} \right] \right\} \\
 &\approx \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \left[6 \frac{\mathbf{a} \cdot \mathbf{r}}{r^2} \frac{\mathbf{r}}{r} - 2 \frac{\mathbf{a}}{r} \right] \\
 &\equiv \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\mathbf{p} \cdot \mathbf{e}_r) \mathbf{e}_r - \mathbf{p}]
 \end{aligned}$$



$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \quad \text{del see appendix H}$$

Taylor's series

$$f(\mathbf{r} + \boldsymbol{\delta}) = f(\mathbf{r}) + \boldsymbol{\delta} \cdot \nabla f(\mathbf{r}) + \frac{1}{2!} (\boldsymbol{\delta} \cdot \nabla)^2 f(\mathbf{r}) + \dots$$

$$\frac{1}{|\mathbf{r} \pm \mathbf{a}|} = \frac{1}{r} \pm \mathbf{a} \cdot \nabla \frac{1}{r} + \dots = \frac{1}{r} \mp \frac{\mathbf{a} \cdot \mathbf{r}}{r^3} + \dots$$

$$\frac{1}{|\mathbf{r} \pm \mathbf{a}|^3} = \frac{1}{r^3} \pm \mathbf{a} \cdot \nabla \frac{1}{r^3} + \dots = \frac{1}{r^3} \mp 3 \frac{\mathbf{a} \cdot \mathbf{r}}{r^5} + \dots$$

$$\nabla \frac{1}{r^3} = \frac{\partial}{\partial r} \left(\frac{1}{r^3} \right) \nabla r, \nabla r = \frac{\mathbf{r}}{r}$$

$$\mathbf{E}_{\perp} = -\frac{1}{4\pi\epsilon_0} \frac{\mathbf{p}}{r^3}$$

$$\mathbf{E}_{\parallel} = \frac{1}{4\pi\epsilon_0} \frac{2\mathbf{p}}{r^3}$$

The dipole of a point charge can be defined as

$$\mathbf{p} = q\mathbf{r}$$

coordinate-dependent

In general,

$$\mathbf{p} = \sum_i q_i \mathbf{r}_i$$

The dipole has essential significance, only when net charge is 0.

$$\sum_i q_i = 0$$

Force or torque on charges

$$\mathbf{F} = q\mathbf{E}$$

$$\mathbf{F} = q\mathbf{E} + (-q)\mathbf{E} = \mathbf{0}$$

$$\mathbf{M} = \mathbf{a} \times (q\mathbf{E}) + (-\mathbf{a}) \times (-q\mathbf{E})$$

$$= 2q\mathbf{a} \times \mathbf{E}$$

$$= \mathbf{p} \times \mathbf{E}$$

the equation of motion is

$$- pE \sin \theta = I \ddot{\theta}$$

The period of small oscillation is

$$T = 2\pi \sqrt{\frac{I}{pE}}$$

*In a non-uniform field

$$\begin{aligned} \mathbf{F} &= -q\mathbf{E}(\mathbf{r} - \mathbf{a}) + q\mathbf{E}(\mathbf{r} + \mathbf{a}) \\ &= -q[\cancel{\mathbf{E}(\mathbf{r})} - \mathbf{a} \cdot \nabla \mathbf{E}(\mathbf{r}) + \dots] + q[\cancel{\mathbf{E}(\mathbf{r})} + \mathbf{a} \cdot \nabla \mathbf{E}(\mathbf{r}) + \dots] \\ &= (\mathbf{p} \cdot \nabla) \mathbf{E}(\mathbf{r}) = \nabla(\mathbf{p} \cdot \mathbf{E}) \end{aligned}$$

$$\begin{aligned} \mathbf{M} &= (\mathbf{r} - \mathbf{a}) \times \mathbf{F}_- + (\mathbf{r} + \mathbf{a}) \times \mathbf{F}_+ \\ &= \mathbf{p} \times \mathbf{E} + \mathbf{r} \times \mathbf{F} \end{aligned}$$

Work-energy relation

$$\begin{aligned} W_{\text{ext}} &= \int_{\theta_0}^{\theta} M_{\text{ext}} \, d\theta \\ &= \int_{\theta_0}^{\theta} pE \sin \theta \, d\theta \\ &= -pE \cos \theta - [-pE \cos \theta_0] \\ &\equiv U(\theta) - U(\theta_0) \end{aligned}$$

$$U = -\mathbf{p} \cdot \mathbf{E} + C$$

For continuous distribution of charge

$$\begin{aligned} \mathbf{E}(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \int \frac{dq(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} (\mathbf{r} - \mathbf{r}') \\ &= \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')d\tau'}{|\mathbf{r} - \mathbf{r}'|^3} (\mathbf{r} - \mathbf{r}') \end{aligned}$$

charge density

charge per unit volume

surface (area) charge density

charge per unit area

linear charge density

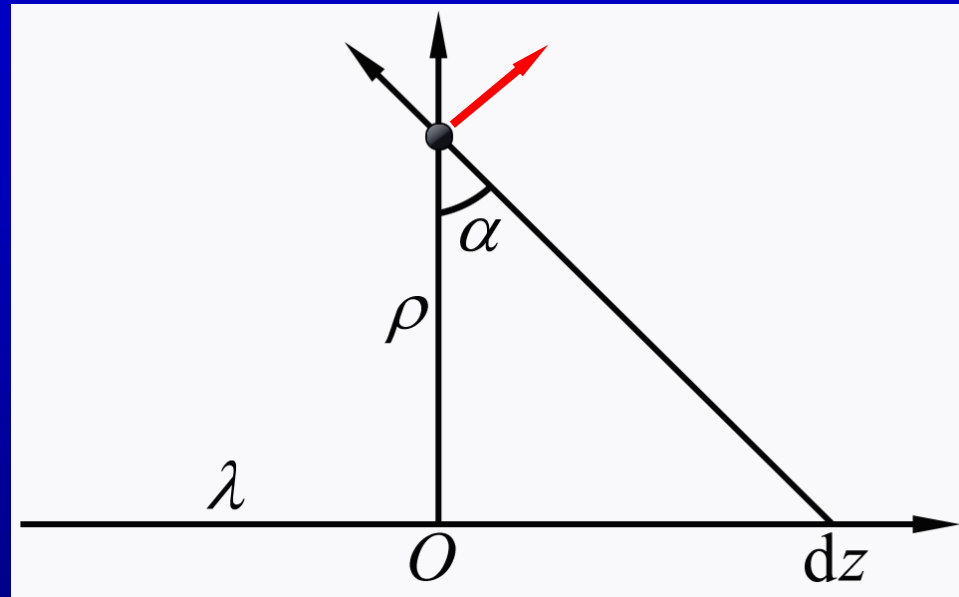
charge per unit length

Example 15.3 Electric field around infinite line of charge with linear charge density λ

$$E = E_{\rho}$$

$$= \int_{-\infty}^{\infty} \frac{1}{4\pi\epsilon_0} \frac{\lambda dz}{r^2} \cos \alpha$$

$$= \int_{-\pi/2}^{\pi/2} \frac{1}{4\pi\epsilon_0} \frac{\lambda \rho \sec^2 \alpha d\alpha}{\rho^2 \sec^2 \alpha} \cos \alpha = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{\rho}$$

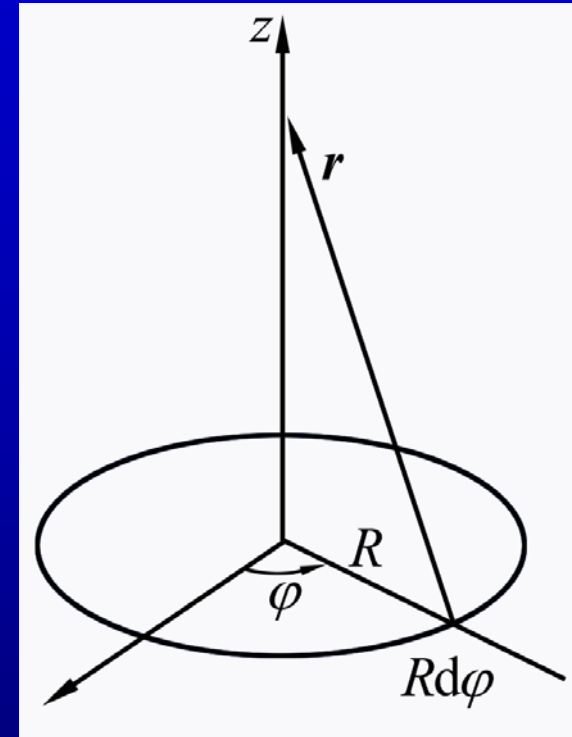


Example 15.4 Electric field around ring of linear charge density $\lambda = q/2\pi R$.

$$\begin{aligned} E = E_z &= \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \frac{q}{2\pi R} \times \frac{R d\phi}{r^2} \times \frac{z}{r} \\ &= \frac{1}{4\pi\epsilon_0} \frac{qz}{r^3} \\ &= \frac{1}{4\pi\epsilon_0} \frac{qz}{(z^2 + R^2)^{3/2}} \end{aligned}$$

For $z \gg R$,

$$E \rightarrow \frac{1}{4\pi\epsilon_0} \frac{q}{z^2} \quad (\text{a point charge.})$$



Extend to a charge on a disc

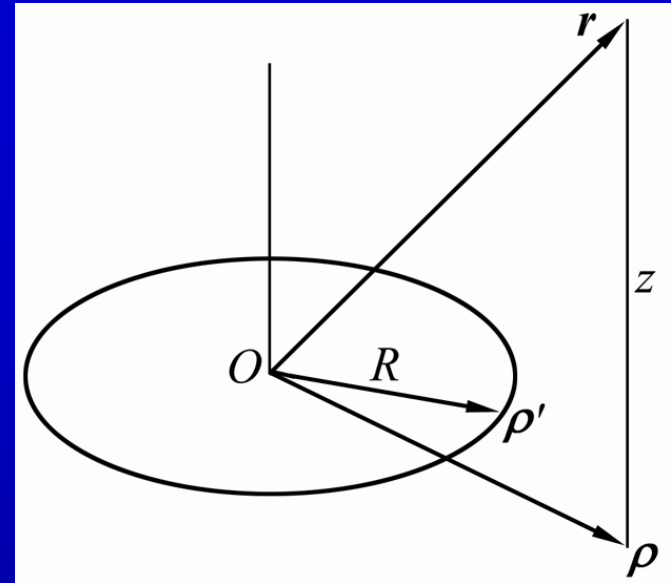
$$\begin{aligned} E_z &= \frac{1}{4\pi\epsilon_0} \int_0^R \frac{2\pi\rho \, d\rho \, \sigma \, z}{(z^2 + \rho^2)^{3/2}} \\ &= \frac{1}{4\pi\epsilon_0} 2\pi \sigma z \left[-\frac{1}{(z^2 + \rho^2)^{1/2}} \right]_0^R \\ &= \frac{1}{4\pi\epsilon_0} 2\pi \sigma \left[\operatorname{sgn} z - \frac{z}{(z^2 + R^2)^{1/2}} \right] \end{aligned}$$

For $R \rightarrow \infty$,

$$E = \frac{1}{4\pi\epsilon_0} 2\pi\sigma \operatorname{sgn} z$$

σ – surface charge density

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \frac{\mathbf{r} - \boldsymbol{\rho}'}{|\mathbf{r} - \boldsymbol{\rho}'|^3} \frac{q}{2\pi R} R d\varphi$$



$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \times \left\{ \mathbf{e}_r + \frac{3}{4} \frac{R^2}{r^2} \left[\left(5 \frac{\rho^2}{r^2} - 2 \right) \mathbf{e}_r - 2 \frac{\rho}{r} \mathbf{e}_\rho \right] + \dots \right\}$$

monopole

quadrupole

Assignment: 15.1, 15.3, 15.6

15.3 Gauss' law

Flux of a vector field

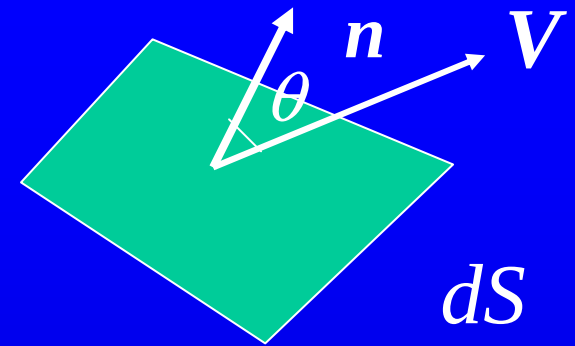
vector field $V(\mathbf{r})$

dS - an infinitesimal area on the surface S

$\mathbf{n}$ -normal vector (unit vector perpendicular to the surface)

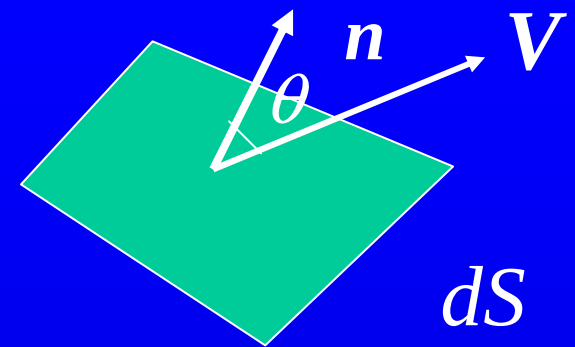
Flux through dS

$$d\phi = V dS \cos \theta$$
$$= \mathbf{V} \cdot \mathbf{n} dS = \mathbf{V} \cdot d\mathbf{S}$$



$$\theta < \frac{\pi}{2}, \quad d\phi > 0 \quad \text{outgoing}$$

$$\theta > \frac{\pi}{2}, \quad d\phi < 0 \quad \text{incoming}$$



Flux through the surface S is

$$\begin{aligned}\phi &= \int V dS \cos \theta \\ &= \int \mathbf{V} \cdot \mathbf{n} dS = \int \mathbf{V} \cdot d\mathbf{S}\end{aligned}$$

Example: Flux of particles

velocity field $\mathbf{v}$

density of particles n (number of particles per unit volume)

The number of particles passing through dS per unit time $nv \cos \theta dS = n\mathbf{v} \cdot \mathbf{n} dS$

The total flux of particle through a surface S

$$\phi = \int_S n\mathbf{v} \cdot \mathbf{n} dS.$$

Assume each particle carries a charge q . The total charge passing through the surface S per unit time(intensity of the electric current) is

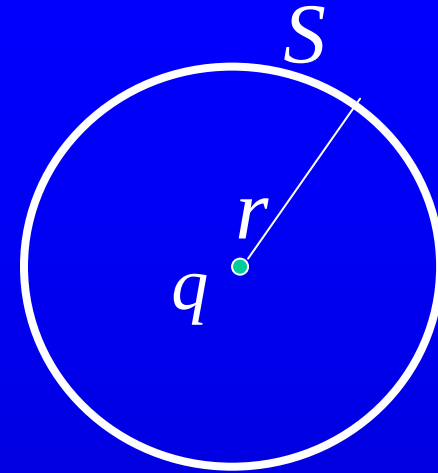
$$I = \int_S nq\mathbf{v} \cdot \mathbf{n}dS = \int_S \mathbf{j} \cdot \mathbf{n}dS,$$

where $\mathbf{j} = nq\mathbf{v}$ is the electric current density.

Gauss' law for electric field

S —spherical surface
concentric with q

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}}}{r^2}$$



$\hat{\mathbf{r}}$ — unit vector along the radial direction

The electric flux through the spherical surface is

$$\Phi_E = \oint \mathbf{E} \cdot d\mathbf{S} = \oint \mathbf{E} \cdot \mathbf{n} dS$$

$$\because \mathbf{n} = \hat{\mathbf{r}}$$

$$\therefore \Phi_E = \frac{q}{4\pi\epsilon_0 r^2} \oint dS = \frac{q}{4\pi\epsilon_0 r^2} (4\pi r^2) = \frac{q}{\epsilon_0}$$

S —arbitrary closed surface

a. q is inside S .

The flux passing through dS :

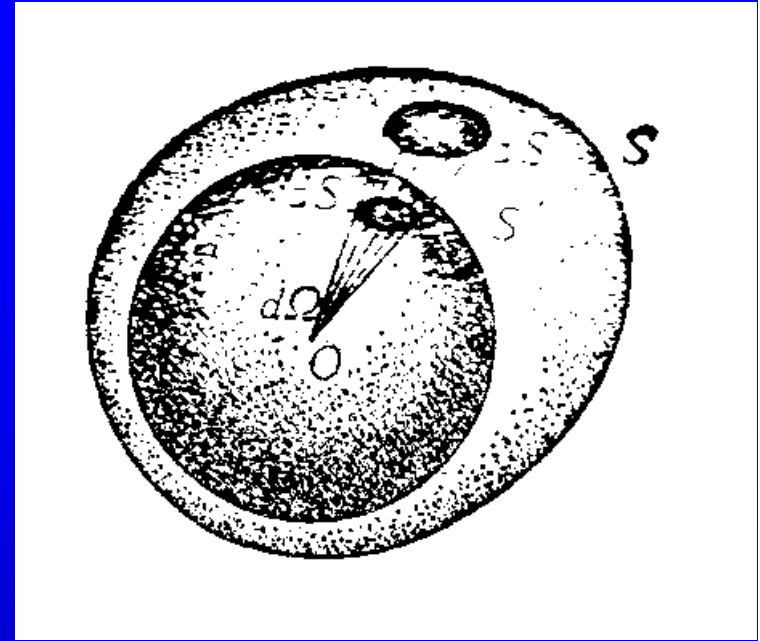
$$d\Phi_E = \mathbf{E} \cdot \mathbf{n} dS = E dS \cos \theta$$

$$= \frac{q}{4\pi\epsilon_0 r^2} \cos \theta dS = \frac{q}{4\pi\epsilon_0} \frac{\cos \theta dS}{r^2}$$

$$\frac{q}{4\pi\epsilon_0} d\Omega$$

$d\Omega$ —solid angle subtended by dS as viewed from q .

$$\Phi_E = \oint \mathbf{E} \cdot \mathbf{n} dS = \frac{q}{4\pi\epsilon_0} \oint d\Omega = \frac{q}{\epsilon_0}$$



b. q is outside S .

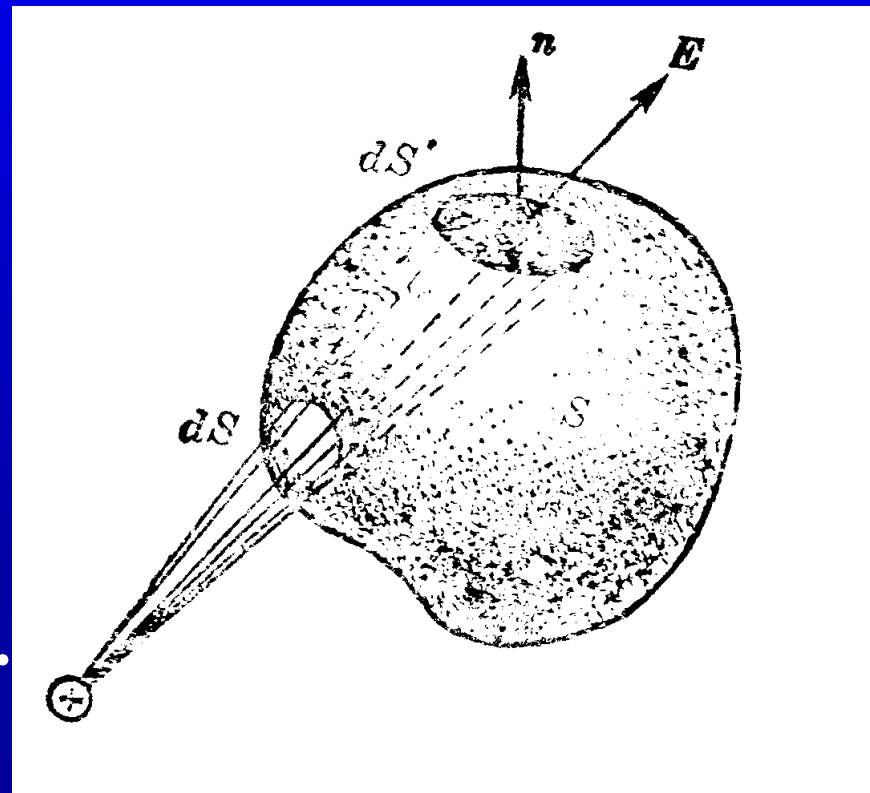
Because the subtended solid angles for dS' and dS are exactly the same, the flux through dS' and dS are equal in magnitude but opposite in sign.

They cancel each other.

$$\Phi_E = 0.$$

In summary:

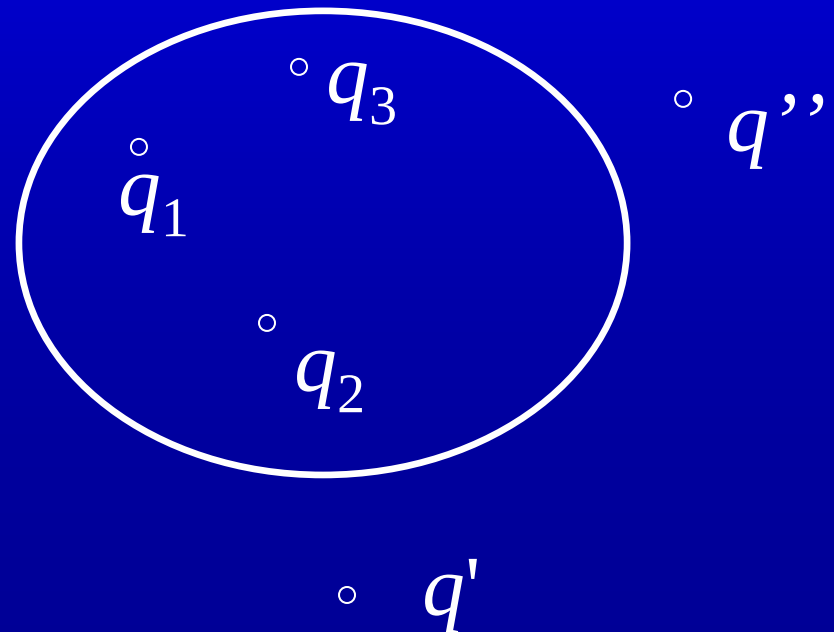
$$\Phi_E = \begin{cases} \frac{q}{\epsilon_0} & q \text{ inside } S; \\ 0 & q \text{ outside } S. \end{cases}$$



Gauss' law The electric flux through a closed surface surrounding charges $q_1, q_2, q_3 \cdots$ is

$$\Phi_E = \oint \mathbf{E} \cdot d\mathbf{S} = \frac{q_{in}}{\epsilon_0},$$

where $q_{in} = \sum q_i$ is the total net charge inside the closed surface.



Solid angle —the space inside a conical surface.

$$\Omega = \frac{S}{R^2}$$

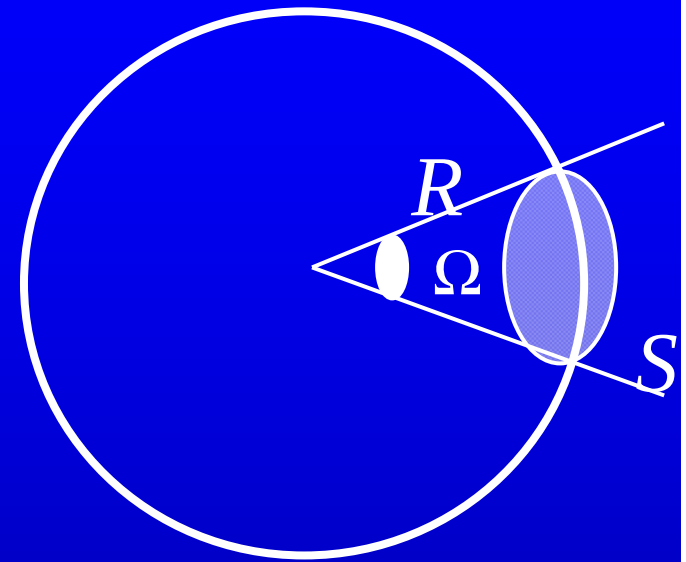
unit: steradian (abbreviation sr)

S —the area on the sphere intercepted by the conical surface.

R —the radius of the sphere.

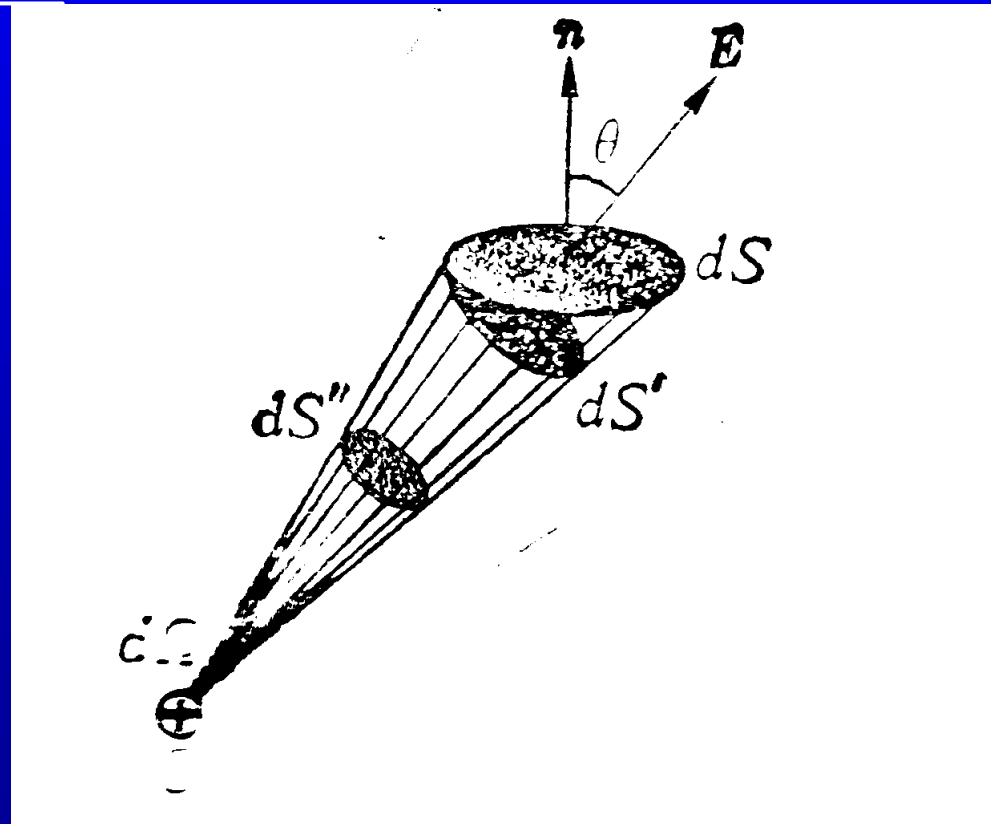
The complete solid angle around a point is

$$\Omega = \frac{4\pi R^2}{R^2} = 4\pi$$



The solid angle subtended by the surface element dS is

$$d\Omega = \frac{dS'}{R^2} = \frac{dS \cos \theta}{R^2}$$

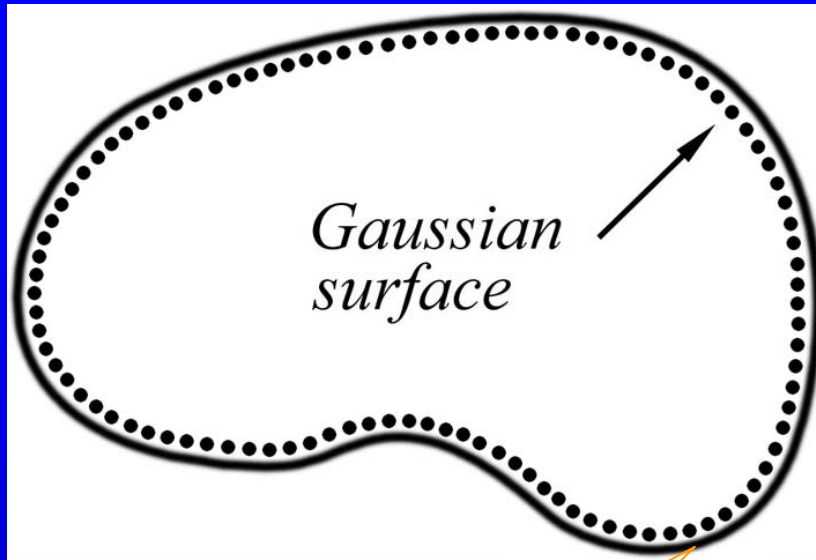


For a point charge, due to symmetry, a spherical *Gaussian surface* centered on the charge is appropriate. Using Gauss' law

$$\oint \mathbf{E} \cdot d\mathbf{S} = E_r \cdot 4\pi r^2 = \frac{q}{\epsilon_0}$$

the electric field is:

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}.$$



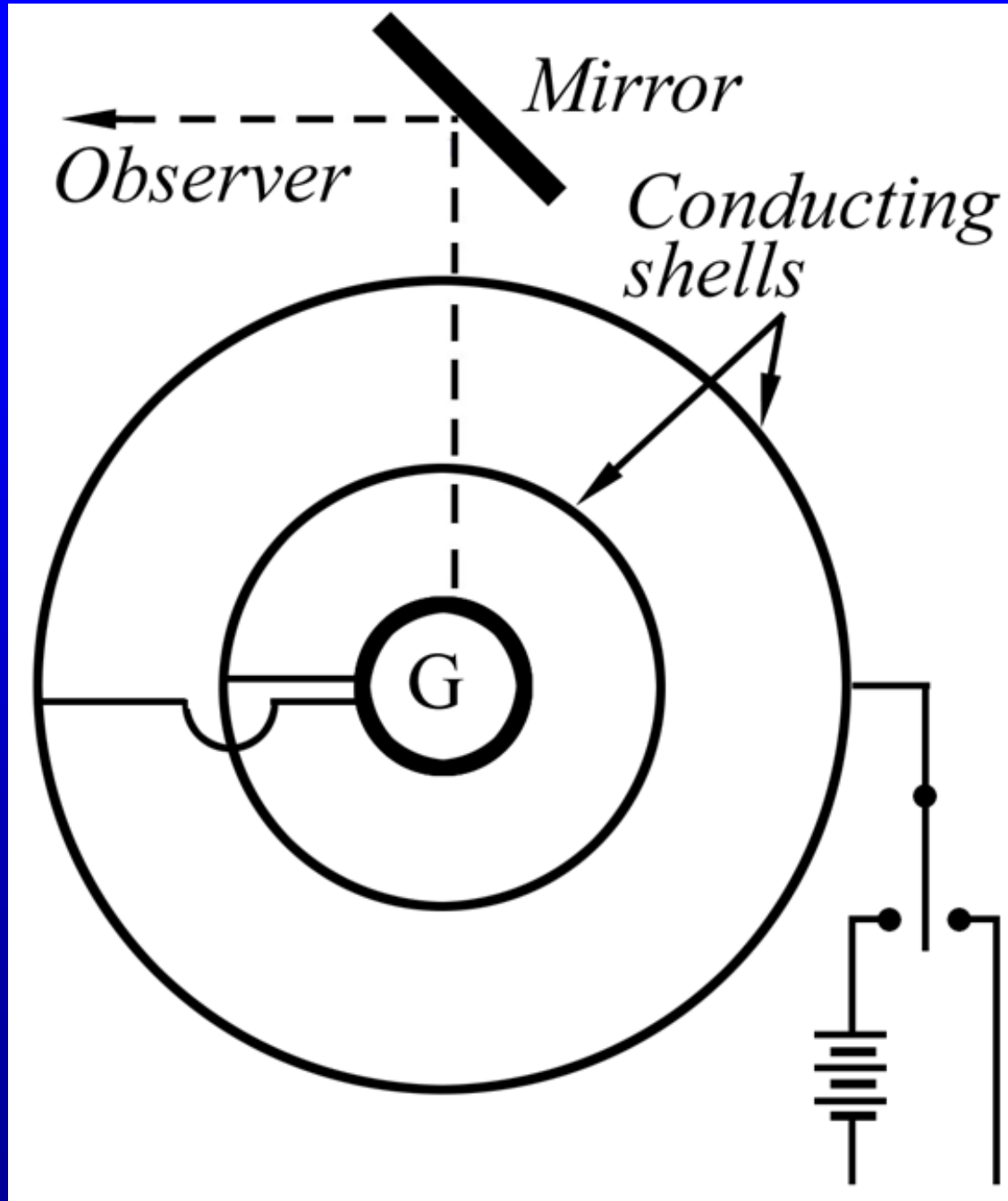
In an insulated conductor:

Charge $\rightarrow$ field $\rightarrow$ motion
 $\rightarrow$ equilibrium $\rightarrow E_{\text{in}} = 0$,
Gauss' law $\rightarrow q_{\text{in}} = 0$

Electric charge can only stay on
the surface of the conductor

conductor

Test Gauss' law and
then Coulomb's law



G-galvanometer

$$\epsilon \leq (2.7 \pm 3.1) \times 10^{-16}$$

Example 15.5 Field of an infinite line of charge of linear density λ .



$$\varepsilon_0 \oint \mathbf{E} \cdot d\mathbf{S} = \varepsilon_0 E_\rho 2\pi\rho h$$

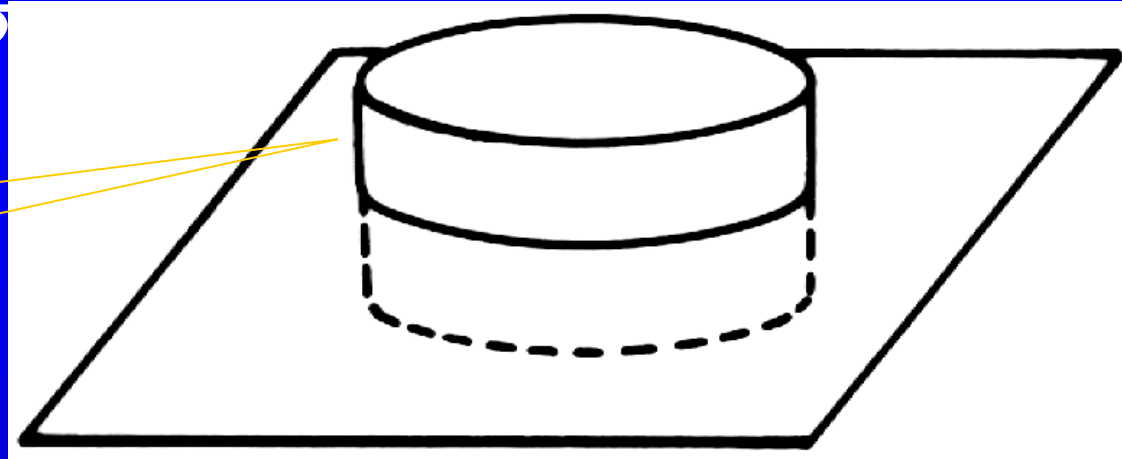
$$q_{\text{in}} = \lambda h$$

$$E_\rho = \frac{1}{4\pi\varepsilon_0} \frac{2\lambda}{\rho}$$

Gaussian surface

Example 15.6 Field of an infinite sheet of charge of surface density σ

Gaussian surface



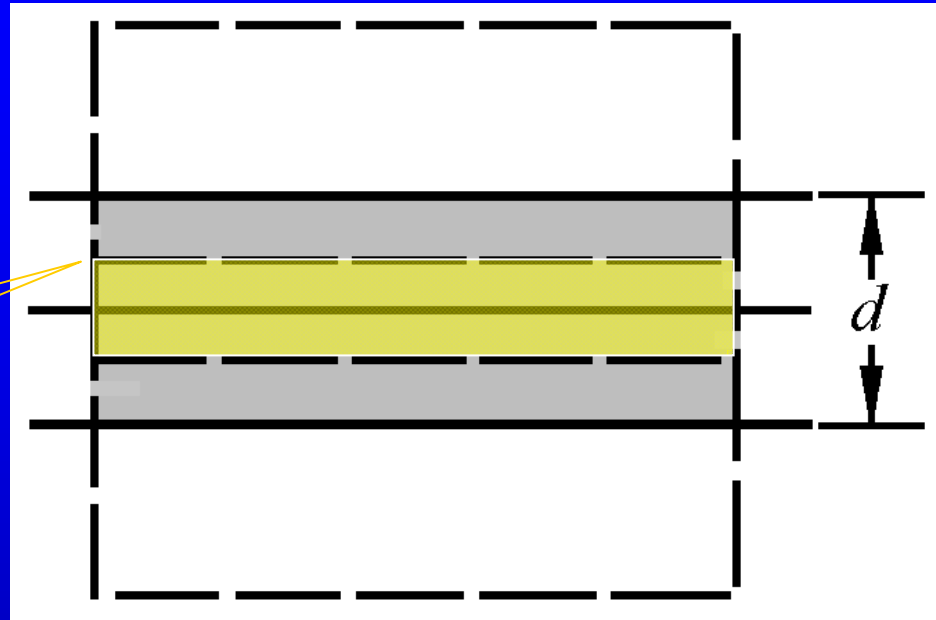
$$\epsilon_0 \oint \mathbf{E} \cdot d\mathbf{S} = \epsilon_0 (EA + EA)$$

$$q_{\text{in}} = \sigma A$$

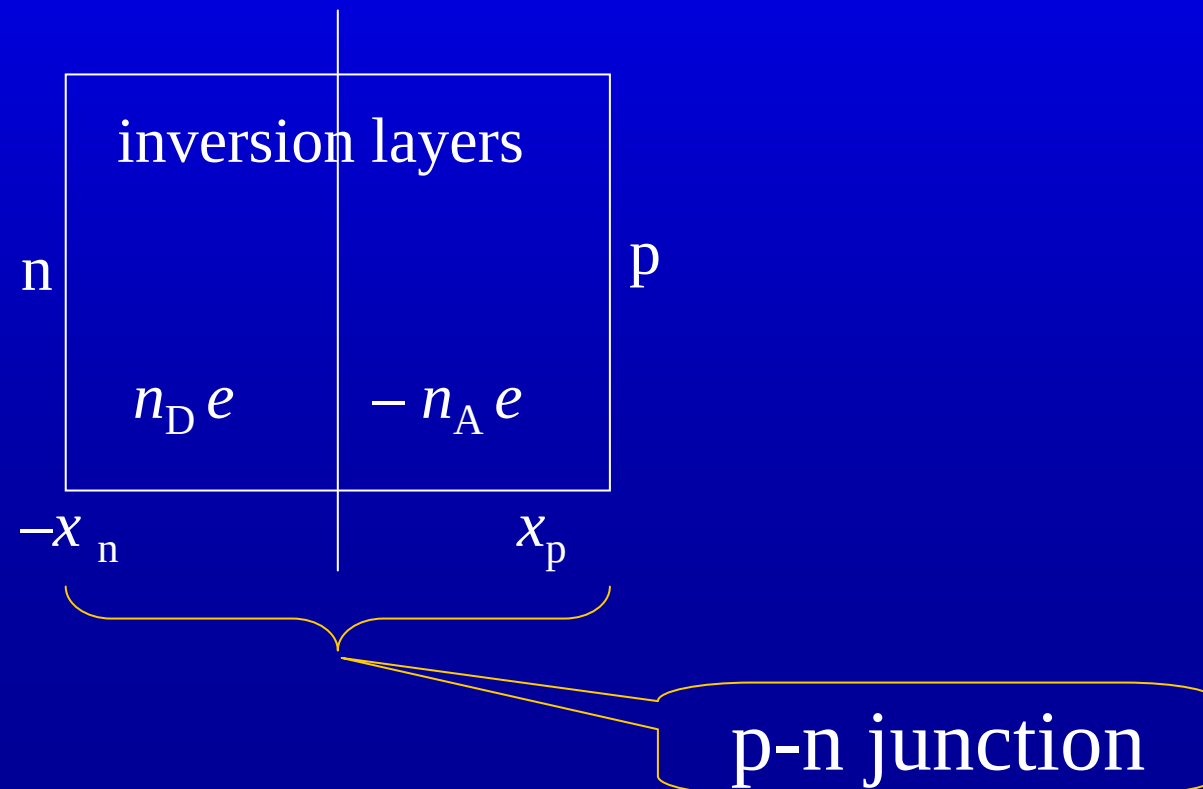
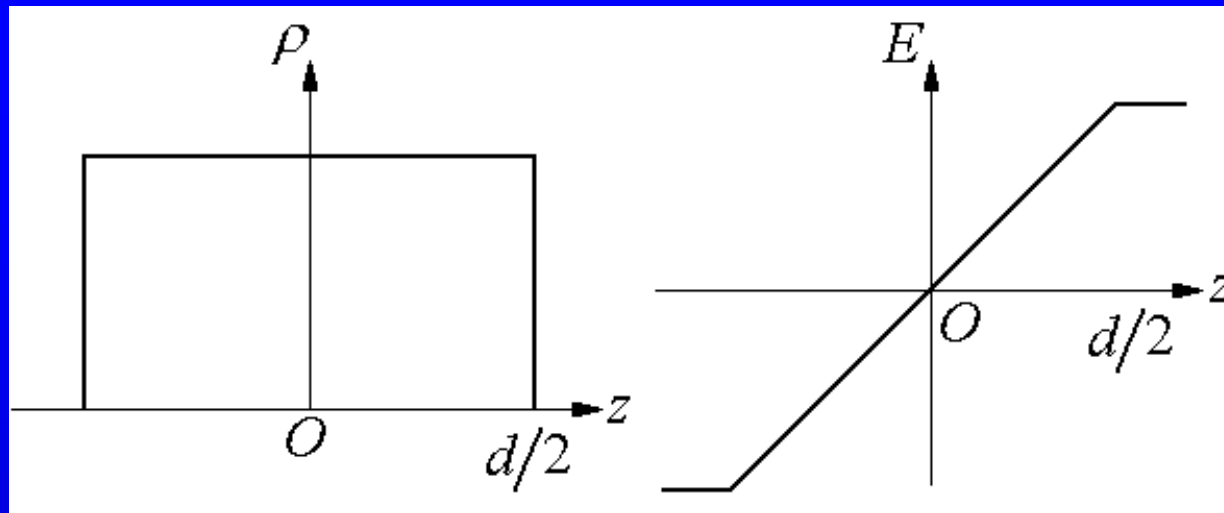
$$E = \frac{\sigma}{2\epsilon_0} = \frac{1}{4\pi\epsilon_0} 2\pi\sigma$$

Example 15.7 Field of infinite sheet of charge with thickness d .

Gaussian surface



$$z < d/2, \quad \epsilon_0(EA + EA) = \rho A 2z, \quad E = \frac{\rho d}{2\epsilon_0} \left(\frac{z}{d/2} \right)$$



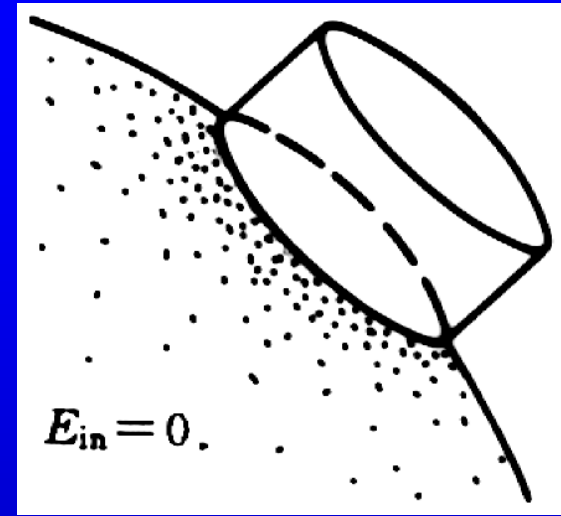
Example 15.8 Electric field near conductor surface of surface density σ

$$\Phi = EA + 0$$

$$q_{\text{in}} = \sigma A$$

$$\epsilon_0 EA = \sigma A$$

$$E = \frac{\sigma}{\epsilon_0}$$



Example 15.9 Typically the earth has a small E
 $=130\text{N/C}$ downward

$$Q = \varepsilon_0 \oint \mathbf{E} \cdot d\mathbf{S} = -\varepsilon_0 E_r 4\pi R_{\oplus}^2$$

$$Q = -5.92 \times 10^5 \text{ C}$$

$$\sigma = \frac{Q}{4\pi R_{\oplus}^2}$$

$$= -\varepsilon_0 E_r = -\frac{1}{4\pi \times 9 \times 10^9} \times 130 \text{ C/m}^2$$

$$= -1.15 \times 10^{-9} \text{ C/m}^2$$

Example 15.10 **Dielectric strength** is the minimum field strength to breakdown material. For dry air, it is

$$E_b \geq 3 \times 10^6 \text{ N/C}$$

$$\frac{Q}{4\pi\epsilon_0 R^2} \leq E_b$$

When $R = 10^{-2}\text{m}$, $Q = 3.3 \times 10^{-8}\text{C}$.

While $R = 1\text{m}$, $Q = 3.3 \times 10^{-4}\text{C}$

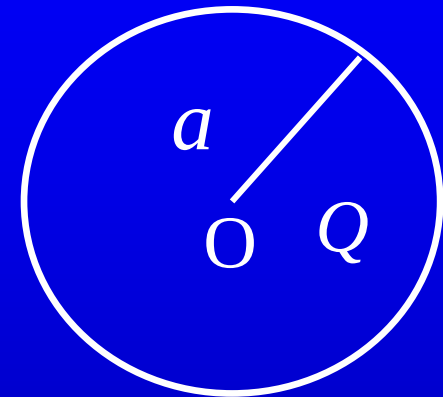
Example Using Gauss' law, determine the electric field of a uniformly charged sphere.

Solution: Because of the symmetry, the orientation of the electric field must be radial. Therefore, making a spherical Gaussian surface of radius r concentric with the charged sphere, the flux through it is

$$\Phi_E = E \cdot 4\pi r^2.$$

For $r > a$, the charge inside the Gaussian surface is Q . Thus, from Gauss' law, we have

$$E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0} \quad \text{or} \quad E = \frac{Q}{4\pi\epsilon_0 r^2}.$$

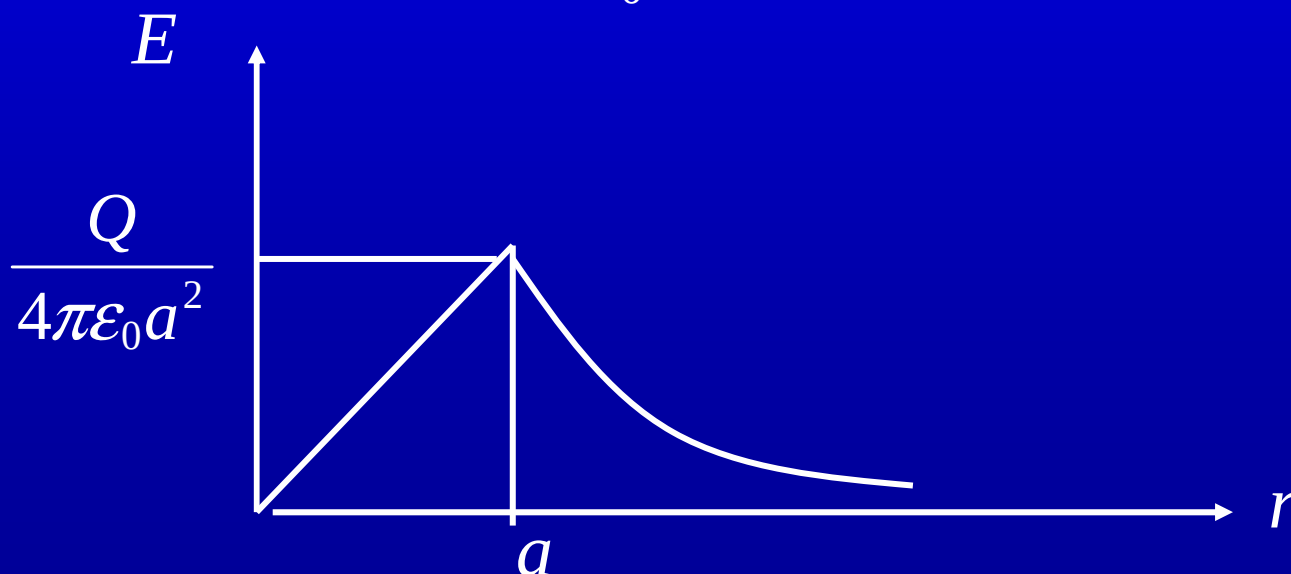


For $r < a$, the charge inside the Gaussian surface is

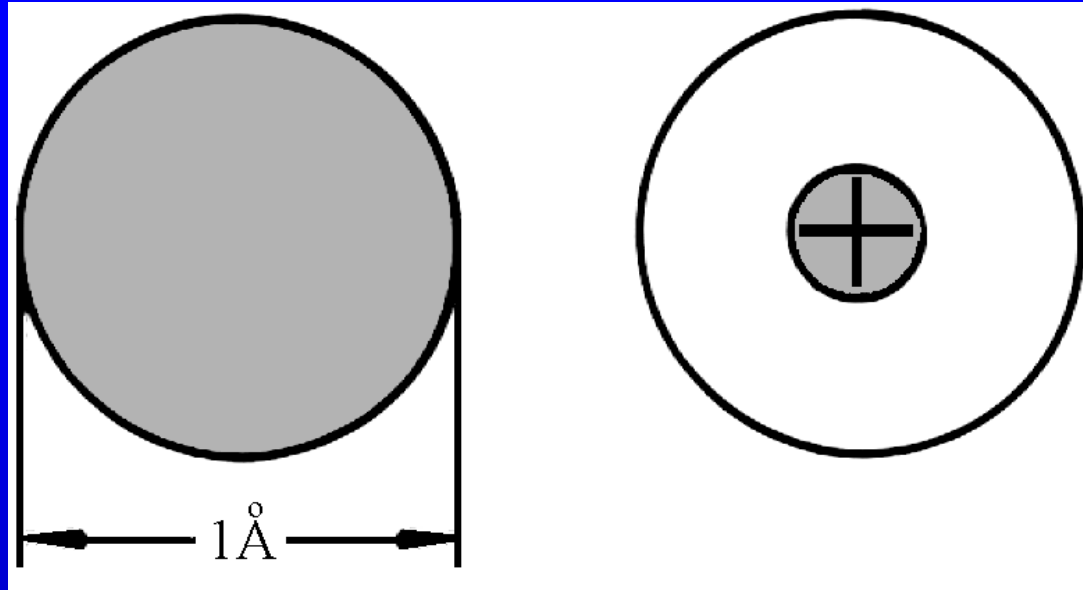
$$Q' = \frac{Q}{4\pi a^3 / 3} (4\pi r^3 / 3) = \frac{Qr^3}{a^3}.$$

Therefore, Gauss' law gives

$$E \cdot 4\pi r^2 = \frac{Qr^3}{\epsilon_0 a^3} \quad \text{or} \quad E = \frac{Qr}{4\pi\epsilon_0 a^3}.$$



atomic models



Thomson model (plum pudding model)

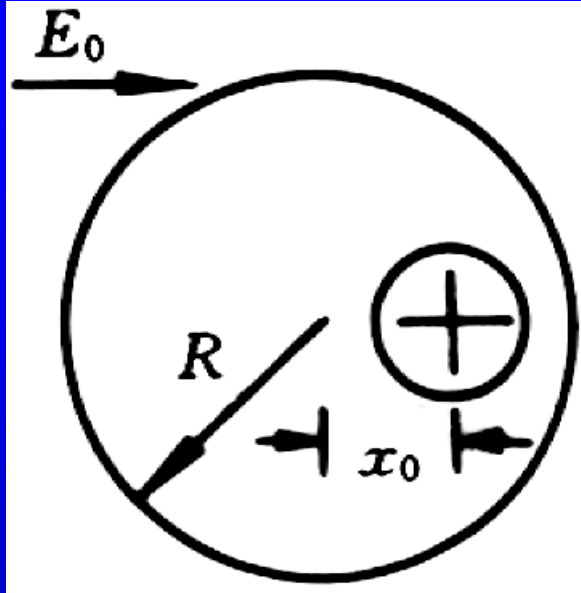
$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \sim 4.4 \times 10^{13} \text{ N/C}$$

Rutherford nuclear atom model

$$E = 2.3 \times 10^{21} \text{ N/C}$$

“fired a 15-inch shell at a piece of tissue paper and it came back and hit you”

Example 15.11 Induced dipole of an atom in E_0



$$E_{\text{core}} = E_0 + \frac{1}{4\pi\epsilon_0} \frac{(-Ze)x_0}{R^3} = 0$$

$$x_0 = 4\pi\epsilon_0 \frac{E_0 R^3}{Ze}$$

Induced dipole is

$$p = Ze x_0 = 4\pi\epsilon_0 E_0 R^3$$

Assignment: 15.9; 15.12

*Gauss' law in differential form

S is an arbitrary closed surface and V is the volume inside it. For a continuous distribution of electric charge, Gauss' law reads



$$\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\epsilon_0} \int_V \rho d\tau.$$

Using Gauss' theorem (divergence theorem)

$$\oint_S \mathbf{E} \cdot d\mathbf{S} = \int_V \text{div } \mathbf{E} d\tau = \int_V \nabla \cdot \mathbf{E} d\tau,$$

we have

$$\int_V \nabla \cdot \mathbf{E} d\tau = \int_V \frac{\rho}{\epsilon_0} d\tau.$$

divergence of $\mathbf{E}$

V —arbitrary $\longrightarrow$ the integrands must equal

$$\text{div } \mathbf{E} = \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}.$$

15.4 Electric potential

gradV gradient of V

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}}{r^3} = \frac{q}{4\pi\epsilon_0} \left(-\nabla \frac{1}{r} \right) = -\nabla \left(\frac{q}{4\pi\epsilon_0} \frac{1}{r} \right) \equiv -\nabla V.$$

$$V = \frac{q}{4\pi\epsilon_0 r} + C.$$

$$\begin{aligned} \nabla \frac{1}{r} &= \mathbf{i} \frac{\partial}{\partial x} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) + \dots \\ &= \mathbf{i} \left(-\frac{1}{2} \frac{2x}{(x^2 + y^2 + z^2)^{3/2}} \right) + \dots \\ &= -\mathbf{i} \frac{x}{r^3} - \mathbf{j} \frac{y}{r^3} - \mathbf{k} \frac{z}{r^3} = -\frac{\mathbf{r}}{r^3}. \end{aligned}$$

$$\nabla V \cdot \mathbf{dl} = \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz = dV \quad \text{---total differential}$$

path independent

$$\int_A^B \mathbf{E} \cdot \mathbf{dl} = \int_A^B -\nabla V \cdot \mathbf{dl} = \int_A^B -dV = -(V_B - V_A).$$

According to the principle of superposition, the electric field produced by a set of point charges is

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 \cdots$$

where $\mathbf{E}_i$ the field produced by q_i . Then

$$\int_A^B \mathbf{E} \cdot d\mathbf{l} = \int_A^B \mathbf{E}_1 \cdot d\mathbf{l} + \int_A^B \mathbf{E}_2 \cdot d\mathbf{l} + \int_A^B \mathbf{E}_3 \cdot d\mathbf{l} + \cdots$$

is also path independent.

$\mathbf{E}$ —the electric force acted on a unit charge

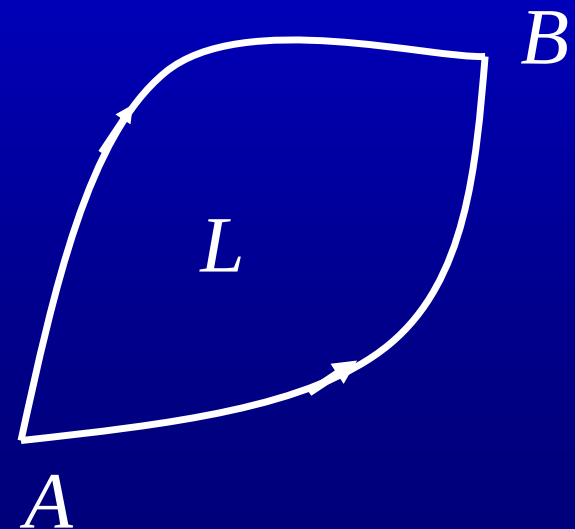
$\int_A^B \mathbf{E} \cdot d\mathbf{l}$ —the work done by the electric field on a unit charge

Conclusion: When a charge is moved in the static electric field, the work done by the field is independent of the path. The static electric field corresponds to a conservative force.

$\int_A^B \mathbf{E} \cdot d\mathbf{l}$ is path independent.

$\oint_L \mathbf{E} \cdot d\mathbf{l} = 0.$

Theorem of circulation



Definition the electric potential

P_0 —reference point, the potential energy at P_0 is zero. an arbitrary path

The potential energy of a unit charge at point P is

$$V_P = -\int_{P_0}^P \mathbf{E} \cdot d\mathbf{l}.$$

Electric potential

unit: joules/coulomb(J/C) volt(V)



Example Find the electric potential for a single point charge q .

Solution: The electric field around a single point charge is

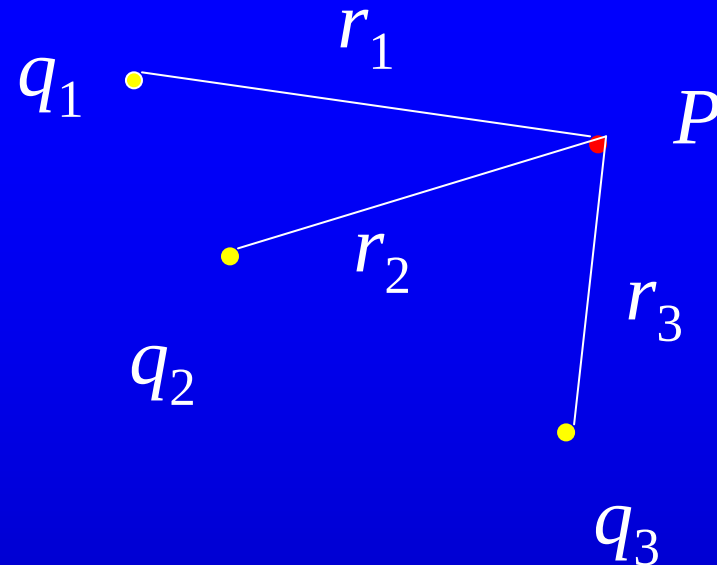
$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}}{r^3} = -\nabla \left(\frac{q}{4\pi\epsilon_0} \frac{1}{r} \right).$$

We generally define $V=0$ at infinity. Then

$$\begin{aligned} V(r) &= -\int_{\infty}^r \mathbf{E} \cdot d\mathbf{l} = \int_{\infty}^r \nabla \left(\frac{q}{4\pi\epsilon_0} \frac{1}{r} \right) \cdot d\mathbf{l} = \int_{\infty}^r d \left(\frac{q}{4\pi\epsilon_0} \frac{1}{r} \right) \\ &= \left. \frac{q}{4\pi\epsilon_0} \frac{1}{r} \right|_{\infty}^r = \frac{q}{4\pi\epsilon_0} \frac{1}{r}. \end{aligned}$$

The electric potential produced by point charges $q_1, q_2, q_3, \dots$ at point P is

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}.$$



For a continuous distribution of charge, we have

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}') d\tau'}{|\mathbf{r} - \mathbf{r}'|} + C$$

The gradient of the electric potential

The electric potential difference between points A and B is

$$V_B - V_A = -\int_A^B \mathbf{E} \cdot d\mathbf{l} = \int_A^B dV$$

where $dV = \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz = \nabla V \cdot d\mathbf{l}.$

Therefore $\mathbf{E} = -\nabla V = -\text{grad } V.$

The electric field is the negative of the gradient of the electric potential.

For three components

$$E_x = -\frac{\partial V}{\partial x}, \dots$$

unit of the electric field: volts/meter (V/m)

Dipole problem

Odd over r & a

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\mathbf{r} - \mathbf{a}|} + \frac{(-q)}{|\mathbf{r} + \mathbf{a}|} \right]$$
$$= \frac{1}{4\pi\epsilon_0} \frac{q}{r} \left[\frac{r}{|\mathbf{r} - \mathbf{a}|} - \frac{r}{|\mathbf{r} + \mathbf{a}|} \right]$$

$$\frac{r}{|\mathbf{r} \mp \mathbf{a}|} = 1 \pm \frac{\mathbf{a} \cdot \mathbf{r}}{r^2} + \dots$$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \frac{2\mathbf{a} \cdot \mathbf{r}}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \mathbf{r}}{r^3}$$

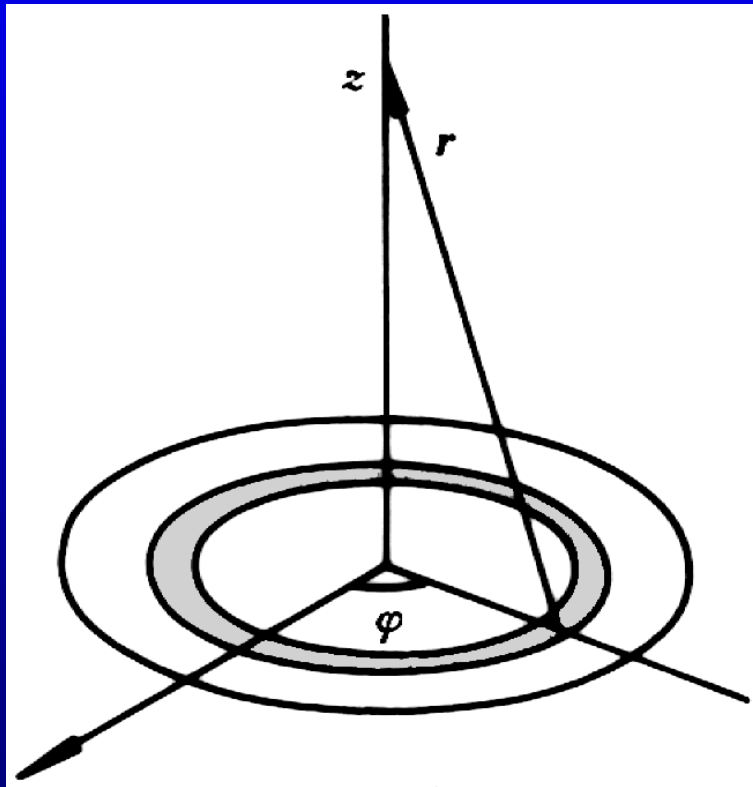
$$\mathbf{E} = -\nabla V = -\frac{1}{4\pi\epsilon_0} \left[\frac{\nabla(\mathbf{p} \cdot \mathbf{r})}{r^3} + (\mathbf{p} \cdot \mathbf{r}) \nabla \frac{1}{r^3} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\mathbf{p} \cdot \mathbf{e}_r) \mathbf{e}_r - \mathbf{p}]$$

$$\nabla(\mathbf{p} \cdot \mathbf{r}) = \mathbf{i} \frac{\partial}{\partial x} (p_x x + p_y y + p_z z) + \dots = \mathbf{i} p_x + \dots = \mathbf{p}.$$

$$\nabla \frac{1}{r^3} = -\frac{3}{r^4} \nabla r = -3 \frac{\mathbf{r}}{r^5}$$

Example 15.12 Find the potential along axis of a uniformly charged disc of radius R and surface density σ .



$$V(z) = \frac{1}{4\pi\epsilon_0} \int_0^R \frac{\sigma 2\pi\rho \, d\rho}{(z^2 + \rho^2)^{1/2}}$$

$$= \frac{1}{4\pi\epsilon_0} 2\pi\sigma \left[(z^2 + R^2)^{1/2} - |z| \right]$$

$$= \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{|z|} & z \gg R \\ -\frac{1}{4\pi\epsilon_0} 2\pi\sigma |z| & z \ll R \end{cases}$$

Example 15.13 Find the potential of a uniformly charged infinite sheet.

Solution: From Example 15.6, the electric field is

$$E = \begin{cases} \frac{\sigma}{2\epsilon_0}, z > 0, \\ -\frac{\sigma}{2\epsilon_0}, z < 0. \end{cases}$$

Using the relation $E = -dV/dz$, we have

$$\frac{dV}{dz} = \begin{cases} -\frac{\sigma}{2\epsilon_0}, z > 0, \\ \frac{\sigma}{2\epsilon_0}, z < 0. \end{cases}$$

Taking the potential at sheet as zero, we have

$$V = \begin{cases} -\frac{\sigma}{2\epsilon_0} z, & z > 0, \\ \frac{\sigma}{2\epsilon_0} z, & z < 0, \end{cases} = -\frac{\sigma}{2\epsilon_0} |z|.$$

Equipotential surfaces—surfaces having the same potential at all points.

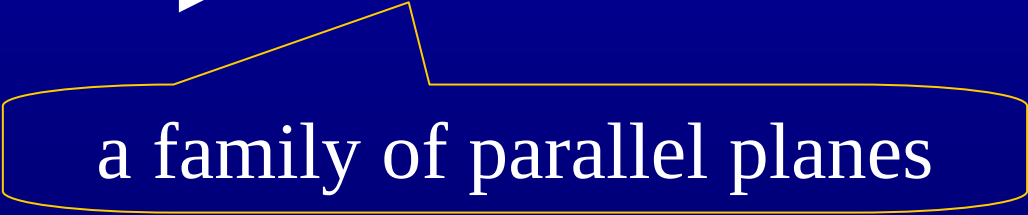
In the case of a single point charge, the equipotential surfaces are

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} = C$$

or $r = C'$.  a family of concentric spheres

For a uniformly charged infinite sheet

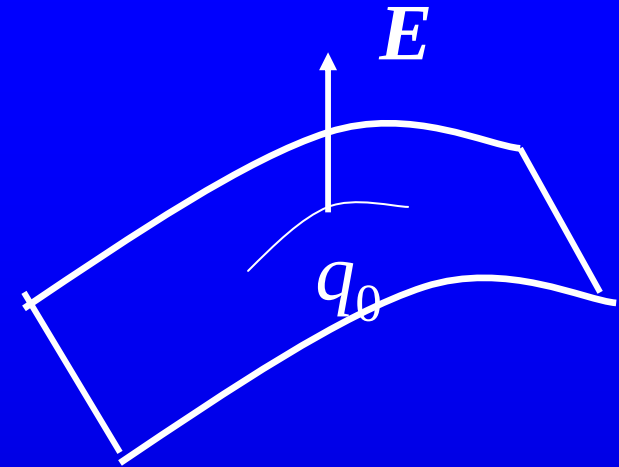
$$V = -\frac{\sigma}{2\epsilon_0} |z| = C. \longrightarrow Z = C'.$$

 a family of parallel planes

If a test charge is moved on an equipotential surface for an arbitrary displacement, no work is done by the field.



E is orthogonal to the equipotential surface at each point.

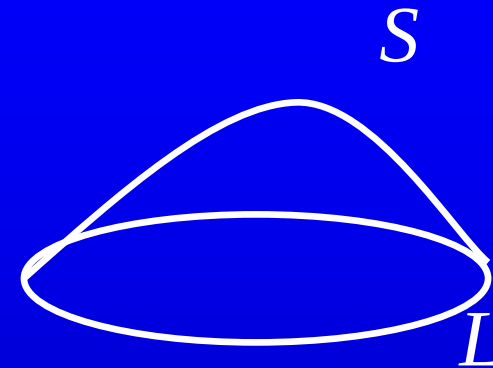


Differential form of the theorem of circulation

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = 0.$$

Stokes' theorem:

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S}.$$



$$\nabla \times \mathbf{E} = 0.$$

rot $\mathbf{E}$, rotation of $\mathbf{E}$

Rotation-free

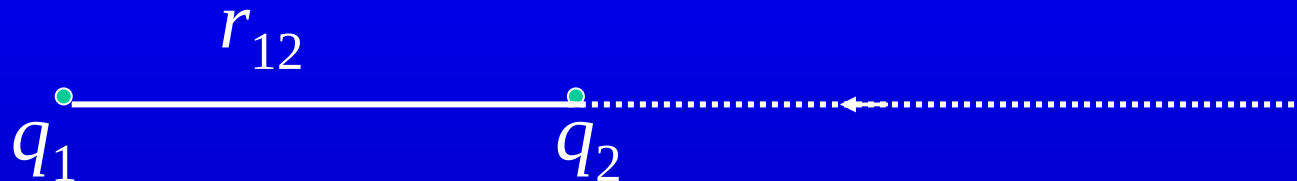
+

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}.$$

with source

15.5 Electric potential energy of charged systems

The electric potential energy of a pair of point charges is just the work done to bring them together.



$$U = W = q_2 V_{12} = q_2 \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_{12}} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

$$V_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_{12}} \quad \text{—the electric potential produced by } q_1 \text{ at the position of } q_2.$$

Two protons (in a nucleus ^{238}U) $6.0 \times 10^{-15} \text{ m}$ apart

$$U = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r} = 2.4 \times 10^5 \text{ eV}$$

$$\gg k_B 300 \text{ K} \approx \frac{1}{40} \text{ eV}$$

Bohr model of hydrogen atom $a_0 = \frac{4\pi\epsilon_0 \hbar^2}{me^2}$

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{a_0^2} = m \frac{v^2}{a_0}, \quad \frac{1}{2} m v^2 = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{e^2}{a_0}, \quad U = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{a_0}$$

$$E = T + U = \frac{1}{2} U = -13.6 \text{ eV}$$

***Electron's speed is**

$$v = \left[\frac{1}{4\pi\epsilon_0} \frac{e^2}{ma_0} \right]^{1/2} = \left[\frac{1}{4\pi\epsilon_0} \frac{e^2}{m \left(\frac{4\pi\epsilon_0 \hbar^2}{me^2} \right)} \right]^{1/2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{e^2}{\hbar c} c \equiv \alpha c$$

$$\alpha = \frac{1}{137}$$

Fine structure constant

For an assembly of n point charges, if we bring them one by one to their positions from the infinity, the works done are

$$q_1: \quad W_1 = 0$$

$$q_2: \quad W_2 = q_2 V_{12}$$

$$q_3: \quad W_3 = q_3 (V_{13} + V_{23})$$

$$\vdots$$

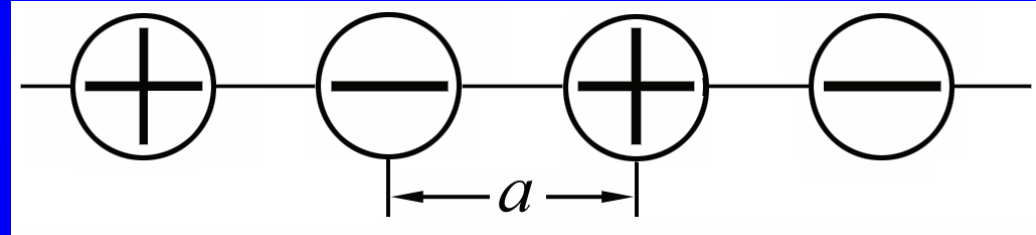
$$q_n: \quad W_n = q_n (V_{1n} + V_{2n} + \cdots + V_{n-1,n}).$$

Therefore, the potential energy of the system is

$$\begin{aligned} U = W &= \sum_{i=1}^n W_i = \sum_{i=1}^n q_i \left(\sum_{j=1}^{i-1} V_{ji} \right) \\ &= \sum_{i=1}^n \sum_{j=1}^{i-1} q_i V_{ji} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j=1}^{i-1} \frac{q_i q_j}{r_{ij}} = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \sum_{\substack{i,j=1 \\ i \neq j}}^n \frac{q_i q_j}{r_{ij}} \\ &= \frac{1}{2} \sum_{\substack{i,j=1 \\ i \neq j}}^n U_{ij} = \frac{1}{2} \sum_{i=1}^n q_i \left(\sum_{j=1, j \neq i}^n \frac{1}{4\pi\epsilon_0} \frac{q_j}{r_{ij}} \right) = \frac{1}{2} \sum_{i=1}^n q_i V_i \end{aligned}$$

V_i is the electric potential on the position of q_i produced by all other charges except q_i itself.

1D ionic crystal



$$U_C = \frac{1}{2} N \sum_{i \neq 1} U_{i1}$$

$$\sum_{i \neq 1} U_{i1} = 2Ze \left(-\frac{1}{4\pi\epsilon_0} \frac{Ze}{a} \right) \left[1 - \frac{1}{2} + \frac{1}{3} - \dots \right]$$

$$= -2 \ln 2 \frac{1}{4\pi\epsilon_0} \frac{(Ze)^2}{a}$$

Madelung constant

binding energy of ionic
crystal

For continuous distribution of charges ρ

$$U = \frac{1}{2} \int V(\mathbf{r}') \rho(\mathbf{r}') d\tau'$$

Self-energy of a spherical charge q is

$$U = \frac{1}{2} \int_0^R \left[\frac{1}{4\pi\epsilon_0} \frac{3q}{2R} \left(1 - \frac{r^2}{3R^2} \right) \right] \rho 4\pi r^2 dr$$

$$= \frac{3}{5} \frac{1}{4\pi\epsilon_0} \frac{q^2}{R}$$

Core-shell

or

$$U = \int V_{<r} \rho(r) 4\pi r^2 dr$$

$$= \frac{1}{4\pi\epsilon_0} \int_0^R \frac{q}{r} \left(\frac{r}{R} \right)^3 \rho 4\pi r^2 dr$$

Suppose electron has structure. Lack of further knowledge about density distribution, one may define a “classical electron radius” by **dimension analysis**

$$\boxed{?} \frac{1}{4\pi\epsilon_0} \frac{e^2}{r_0} \equiv m_0 c^2$$

$$r_0 = \frac{1}{4\pi\epsilon_0} \frac{e^2}{m_0 c^2} = 2.82 \times 10^{-15} \text{ m}$$

Where is U located?

$$U \stackrel{?}{=} \frac{\epsilon_0}{2} \int \mathbf{E} \cdot \mathbf{E} d\tau$$

$$= \frac{\epsilon_0}{2} \int (-\nabla V) \cdot \mathbf{E} d\tau$$

$$= \frac{\epsilon_0}{2} \int [-\nabla \cdot (V\mathbf{E}) + V\nabla \cdot \mathbf{E}] d\tau$$

$$= -\frac{\epsilon_0}{2} \oint V\mathbf{E} \cdot d\mathbf{S} + \frac{1}{2} \int V\rho d\tau$$

$$= \frac{1}{2} \int V\rho d\tau$$

$$u = \frac{\epsilon_0}{2} \mathbf{E} \cdot \mathbf{E}$$

Applying it to point charges,

$$\begin{aligned} U &= \frac{\epsilon_0}{2} \int_0^\infty \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right)^2 4\pi r^2 dr \\ &= \frac{1}{2} \frac{q^2}{4\pi\epsilon_0} \int_0^\infty \frac{1}{r^2} dr \\ &= \frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{q^2}{r} \bigg|_0^\infty \rightarrow \infty \end{aligned}$$

When two point charges exist, the field is a superposition of their fields. The energy of interaction is

$$\begin{aligned}\Delta U &= \frac{\epsilon_0}{2} \int (\mathbf{E}_1 + \mathbf{E}_2) \cdot (\mathbf{E}_1 + \mathbf{E}_2) d\tau - \frac{\epsilon_0}{2} \int E_1^2 d\tau - \frac{\epsilon_0}{2} \int E_2^2 d\tau \\ &= \frac{\epsilon_0}{2} \int (\mathbf{E}_1 \cdot \mathbf{E}_2 + \mathbf{E}_2 \cdot \mathbf{E}_1) d\tau \\ &= \frac{1}{2} \int (\rho_1 V_1 + \rho_2 V_2) d\tau \\ &= \frac{1}{2} (q_1 V_1 + q_2 V_2)\end{aligned}$$

Assignment 15.15, 15.17